

Кобки Гяссона

u, v ом $q_1, \dots, q_n, p_1, \dots, p_n, t$

$$(u, v) = \sum_{i=1}^n \left(\frac{\partial u}{\partial q_i} \frac{\partial v}{\partial p_i} - \frac{\partial u}{\partial p_i} \frac{\partial v}{\partial q_i} \right)$$

Об-ба:

1) $(u, v) = -(v, u)$

2) $(c(t)u, v) = c(t)(u, v)$

3) $(u+v, w) = (u, w) + (v, w)$

4) $\frac{\partial}{\partial t}(u, v) = \left(\frac{\partial u}{\partial t}, v \right) + \left(u, \frac{\partial v}{\partial t} \right)$

5) момента Гяссона

$$((u, v), w) + ((v, w), u) + ((w, u), v) = 0$$

Критерий ПУ

$$\left. \frac{df}{dt} \right|_H = 0$$

$$\left. \frac{df}{dt} \right|_H = \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \dot{q}_i \Big|_H + \frac{\partial f}{\partial p_i} \dot{p}_i \Big|_H \right) + \frac{\partial f}{\partial t} = \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right) + \frac{\partial f}{\partial t}$$

$$\frac{\partial f}{\partial t} + (f, H) = 0 \quad \text{— кр. ПУ}$$

прим.

1) $H = H(p, q)$

$$\frac{\partial H}{\partial t} = 0$$

$$(f, H) = (H, H) = 0$$

H — ПУ

2) q_1 — цик. коорд. ($\Rightarrow \frac{\partial H}{\partial q_1} = 0$)

$$f(p_1) \text{ — ПУ}$$

$$\frac{\partial f}{\partial t} = 0 \quad \frac{\partial f}{\partial q_i} = 0 \quad \frac{\partial f}{\partial p_1} \neq 0 \quad \frac{\partial f}{\partial p_i} = 0 \quad i \neq 1$$

$$(f, H) = - \frac{\partial f}{\partial p_1} \frac{\partial H}{\partial q_1} = 0$$

канон. упр. $f(p_1) = p_1$ — ПУ

3) $H = H(f(q_1, p_1), q_2, p_2, \dots, q_n, p_n, t)$

$$f(q_1, p_1) \text{ — ПУ}$$

$$\frac{\partial f}{\partial t} = 0$$

$$(f, H) \stackrel{?}{=} 0$$

$$(f, H) = \frac{\partial f}{\partial q_1} \frac{\partial H}{\partial p_1} - \frac{\partial f}{\partial p_1} \frac{\partial H}{\partial q_1} = \frac{\partial f}{\partial q_1} \frac{\partial H}{\partial t} \frac{\partial f}{\partial q_1} - \frac{\partial f}{\partial p_1} \frac{\partial H}{\partial t} \frac{\partial f}{\partial p_1} = 0$$

$$H = H(f_1(q_1, p_1), f_2(q_2, p_2), \dots, f_n(q_n, p_n), t)$$

$$f_i(q_i, p_i) = \alpha_i - \Pi U$$

прим.: $H = \underbrace{q_1^2 + p_1^2}_{\text{многомер. осциллятор}} + q_2^2 + p_2^2$

$$4) H = H(f_n(q_n, p_n, f_{n-1}(q_{n-1}, p_{n-1}, \dots), t)$$

$$f_k = f_k(q_k, p_k, f_{k-1}) \quad k = \overline{2, n}$$

$$f_1 = f_1(q_1, p_1)$$

$$f_i - \Pi U$$

$$(f_i, H) = \sum_{k=1}^n (\dots) = \sum_{k=1}^i \left(\frac{\partial f_i}{\partial q_k} \frac{\partial H}{\partial p_k} - \frac{\partial f_i}{\partial p_k} \frac{\partial H}{\partial q_k} \right) = \sum_{k=1}^i \left(\frac{\partial f_i}{\partial f_{i-1}} \frac{\partial f_{i-1}}{\partial f_{i-2}} \dots \frac{\partial f_k}{\partial q_k} \frac{\partial H}{\partial f_n} \frac{\partial f_n}{\partial f_{n-1}} \dots \frac{\partial f_k}{\partial p_k} - \frac{\partial f_i}{\partial f_{i-1}} \dots \frac{\partial f_k}{\partial p_k} \frac{\partial H}{\partial f_n} \frac{\partial f_n}{\partial f_{n-1}} \dots \frac{\partial f_k}{\partial q_k} \right) = 0$$

прим.: $H = q_2^2 + p_2^2(q_1^2 + p_1^2)$

$$5) H = c(t) H_0(q, p)$$

$$H_0 - \Pi U$$

$$\frac{\partial H_0}{\partial t} = 0; \quad (H_0, H) = (H_0, c(t)H_0) = c(t)(H_0, H_0) = 0$$

$$6) H = \frac{\sum_{i=1}^n f_i(q_i, p_i)}{\sum_{i=1}^n g_i(q_i, p_i)} = h$$

$$f_i - h g_i = \alpha_i - \Pi U$$

$$\sum_{i=1}^n f_i - h \sum_{i=1}^n g_i = 0$$

$$f_n - h g_n = - \sum_{i=1}^{n-1} (f_i - h g_i) \quad \text{зависимы!}$$

th Якоби-Пуассона

Если f, g - ΠU с мет. с зам. H , то (f, g) - тоже ΠU

$$\Delta \begin{cases} \frac{\partial f}{\partial t} + (f, H) = 0 \\ \frac{\partial g}{\partial t} + (g, H) = 0 \end{cases}$$

$$\frac{\partial}{\partial t} (f, g) = \left(\frac{\partial f}{\partial t}, g \right) + \left(f, \frac{\partial g}{\partial t} \right) = - \langle (f, H), g \rangle - \langle f, (g, H) \rangle = \langle (H, f), g \rangle + \langle (g, H), f \rangle$$

$$\frac{\partial}{\partial t} (f, g) + \langle (f, g), H \rangle = \langle (H, f), g \rangle + \langle (g, H), f \rangle + \langle (f, g), H \rangle = 0$$

прим.

$$1) q_1 - \text{цикл.} \quad H = (q_2, \dots, q_n, p_1, \dots, p_n) - \Pi U$$

$$p_1 - \Pi U$$

$$(H, p_1) = - \frac{\partial H}{\partial q_1} \frac{\partial p_1}{\partial p_1} = 0$$

0 " 1

2) $\Pi(\vec{r})$ центр. поле ($m=1$)

$$T = \frac{1}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = \frac{1}{2}(p_x^2 + p_y^2 + p_z^2)$$

$$\left(\frac{\partial T}{\partial \dot{x}} = p_x, p_x = \dot{x}\right)$$

$$\vec{K}_0 = \begin{vmatrix} i & j & k \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix} = \begin{bmatrix} p_z y - p_y z \\ p_x z - p_z x \\ p_y x - p_x y \end{bmatrix}$$

$$J_1 = p_z y - p_y z$$

$$J_2 = p_x z - p_z x$$

$$(J_1, J_2) = \frac{\partial J_1}{\partial x} \frac{\partial J_2}{\partial p_x} - \frac{\partial J_1}{\partial p_x} \frac{\partial J_2}{\partial x} + \frac{\partial J_1}{\partial y} \frac{\partial J_2}{\partial p_y} - \frac{\partial J_1}{\partial p_y} \frac{\partial J_2}{\partial y} + \frac{\partial J_1}{\partial z} \frac{\partial J_2}{\partial p_z} - \frac{\partial J_1}{\partial p_z} \frac{\partial J_2}{\partial z} = p_y x - y p_x$$

$$(J_1, J_2) = J_3$$

th условия о разрешимости в квадратурах

Если в кан. сист. $\dot{q} = \frac{\partial H}{\partial p}$, $\dot{p} = -\frac{\partial H}{\partial q}$ есть n первых инт. в инволюции

$$(H_k, k = \overline{1, n}, H_k(\vec{q}, \vec{p}, t))$$

и $\det \left| \frac{\partial H_k}{\partial p_j} \right|_{k,j} \neq 0$, то сист. разреш. в квадратурах.

Условия

H_1, H_2 - в инвол., если $(H_1, H_2) = 0$

прим.

$$H = H(f_1(q_1, p_1), \dots, f_n(q_n, p_n), t)$$

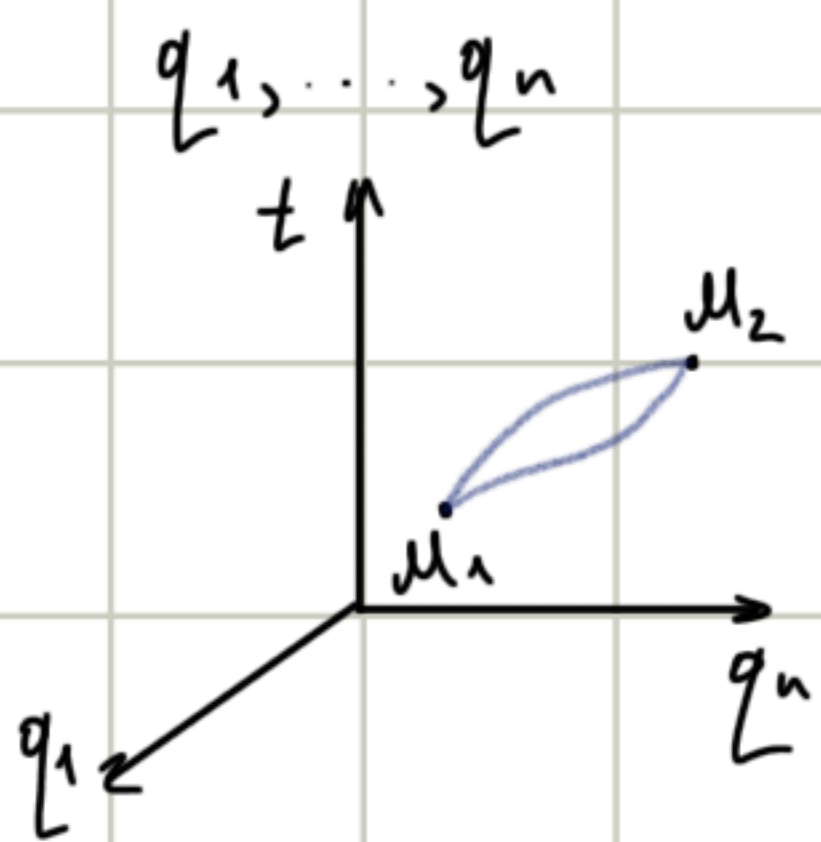
$$f_i(q_i, p_i) = \alpha_i - \pi_i$$

$$\int \frac{\partial f_i}{\partial p_i} \neq 0$$

→ инт. сист.

Принцип Гамильтона

Гамильтонова система с уг. удерж. связями или свобод.

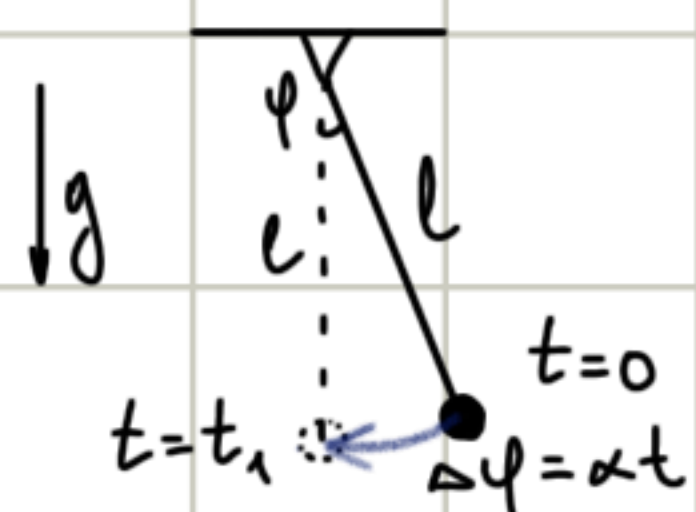


расшир.
ком.
пр-во

$\mathcal{L}$

"прямой" путь - решение краевой задачи с $\mathcal{L}$

"окольный" путь - пути из M_1 в M_2 , допускающие связи, но не прямой



$$\ddot{\varphi} + \frac{g}{l} \varphi = 0$$

φ - прямой путь

$$W = \int_{t_1}^{t_2} L(\bar{q}, \dot{\bar{q}}, t) dt \quad \text{действие по Лагранжу}$$

$\bar{q} = \bar{q}(t)$

Принцип Лагранжа

Действие по Лаг. принимает макс. значения на прямом пути по сравнению с окружающими

$$\Delta. q_i = q_i(t, \alpha) \quad q_i \in (\overline{M_1, M_2}), \quad t \in [t_1, t_2]$$

при $\alpha = 0$ - прямой путь

$$\alpha: -\gamma \leq 0 \leq \gamma$$

$$\bar{q} = \bar{q}(t, \alpha)$$

$$L = L(\bar{q}, \dot{\bar{q}}, t) = L(t, \alpha)$$

$$W = \int_{t_1}^{t_2} L(t, \alpha) dt = W(\alpha)$$



$$\delta W = \frac{\partial W}{\partial \alpha} d\alpha = \int_{t_1}^{t_2} \delta L dt = \int_{t_1}^{t_2} \left[\sum_{i=1}^n \frac{\partial L}{\partial q_i} \delta q_i + \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i \right] dt$$

$$\delta L = \frac{\partial L}{\partial \alpha} d\alpha$$

$$\delta \dot{q}_i = \frac{\partial}{\partial \alpha} \frac{dq_i}{dt} d\alpha = \frac{d}{dt} \frac{\partial q}{\partial \alpha} d\alpha = \frac{d}{dt} \delta q$$

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i dt = \int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_i} \frac{d}{dt} \delta q_i dt = \frac{\partial L}{\partial \dot{q}_i} \delta q_i \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \delta q_i \cdot \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} dt$$

$d(\delta q_i)$

$$\delta q_i(t_1) = \delta q_i(t_2) = 0 \quad \text{из краевой задачи}$$

$$\delta W = \int_{t_1}^{t_2} \sum_{i=1}^n \delta q_i \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right] dt$$

Если путь прямой: $\delta W = 0$

Если $\delta W = 0$, то $\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = 0 \Rightarrow q_i(t, \alpha) - \text{прямой путь}$

Принцип Лагранжа-Остроградского

$$\int_{t_1}^{t_2} (\delta T + \delta A) dt = 0$$

- дП

th Подымова

Если на M_1, M_2 нет сопряж. крив. фокусов для M_1 , то W на прямой пути принимает min значение